



INFRASTRUCTURE

MCP Server Integration Handbook

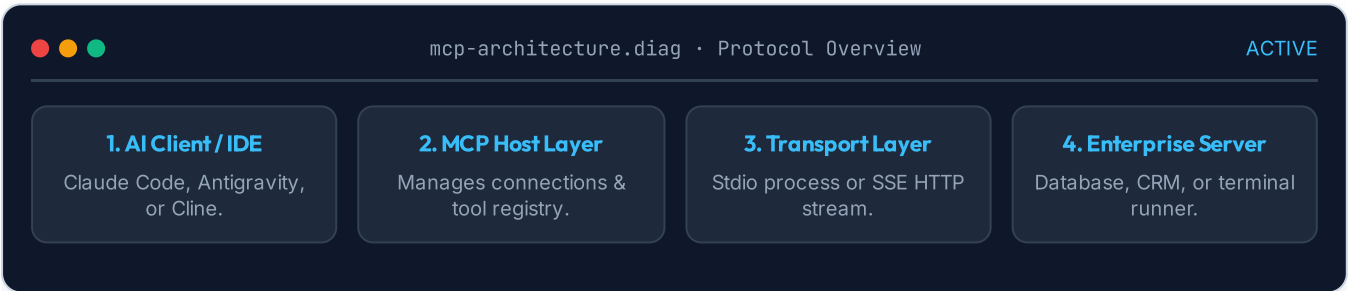
Building and deploying Model Context Protocol (MCP) servers for databases, APIs, and file access.

EXECUTIVE SUMMARY

The Model Context Protocol (MCP) has become the open standard for connecting AI models to enterprise tools and data. This handbook provides end-to-end patterns for creating secure MCP servers with Stdio and SSE transports.

1 Model Context Protocol Architecture

MCP decouples tool implementation from the LLM client. An MCP host coordinates with local or remote servers via JSON-RPC, exposing standardized tools, database queries, and live system resources.



2 Production Implementation & Schema

```
// Production TypeScript MCP Tool Server
server.tool('queryDatabase', { query: z.string() }, async ({ query }) => {
  const result = await db.execute(query);
  return {
    content: [{ type: 'text', text: JSON.stringify(result.rows) }]
  };
});
```



3 Secure Transport & Neon Postgres Integration

Connect MCP servers to serverless relational databases like Neon PostgreSQL, allowing AI agents to inspect schemas, execute migrations, and perform vector searches with strict permission boundaries.

mcp-inspector • Server Diagnostic

CONNECTED

```
[MCP HOST] Connecting to mcp-server-neon via stdio ...
✓ [TOOLS] 8 database tools discovered and registered
✓ [AUTH] Neon project credentials validated successfully
[EXEC] describe_table_schema("users") → 12 columns returned
[SECURITY] Read-only transaction enforcement ACTIVE
```

4 Key Takeaways & Best Practices

- ✓ Transport Selection

Use Stdio transport for local desktop agents and Server-Sent Events (SSE) for distributed cloud teams.
- ✓ Schema Pruning

Only expose the tools an agent actually needs; bloated tool definitions consume valuable context tokens.
- ✓ Read-Only Safety

Enforce read-only database connections by default, requiring explicit user approval for mutations.
- ✓ Error Clarity

Return structured, actionable error strings in tool responses so agents can self-correct when parameters fail.

IMPLEMENTATION SUPPORT

Need engineering assistance integrating these architectures into your business? Contact our team directly at futureai4all@gmail.com.